NORTH CAROLINA AGRICULTURAL AND TECHNICAL STATE UNIVERSITY

MATH 224-Data Analysis Project

Part I: Exploring the Dataset & Identifying Research Questions

AI Tool Usage Instructions

For this project, you are allowed to use one generative AI tool of your choice. Examples include ChatGPT, Microsoft Copilot, or Google Gemini. Our school currently provides free access to Microsoft Copilot, and Gemini offers a free student plan for 12 months — but it is completely up to you which tool you prefer.

Rules for AI Use

- Use only ONE dedicated conversation/chat for this project. All project-related prompts and questions must stay in that same conversation.
- Do not spread project-related questions across multiple chats. If you need help for unrelated study purposes, open a different chat.
- Include your AI interaction link upon submission so your instructor can review your full conversation.
- AI is a tool to assist your thinking — not a replacement for it. You must understand, verify, and take ownership of every answer you submit.

What You SHOULD Do ✓

✓ Good AI Use — Examples

"I have a dataset with variables age, income, and education. Can you explain what summary statistics I should compute and why?"

"My histogram looks right-skewed. What does that mean for interpreting the mean vs. median?"

"Here is my R code — it gives an error. Can you help me find the issue?" [paste your actual code]

"Can you explain what a confidence interval means in simple terms?"

"I got a p-value of 0.03 in my t-test. How do I interpret this in context?"

What You Should NOT Do ✗

✗ Prohibited AI Use — Do NOT do these

Do NOT paste the project question directly and submit whatever the AI writes.
Do NOT ask: "Do Part 1 of my MATH 224 project for me."
Do NOT copy and paste AI output word-for-word without understanding it.
Do NOT use AI to generate R code you cannot explain or did not verify.
Do NOT use multiple chat threads to hide the extent of your AI use.

⚠ Important Reminder

AI tools can make mistakes. Always verify outputs. If AI gives you something that seems off, trust your judgment and ask your instructor.

Reflection Requirement

At the end of your submission, include a short paragraph (5–8 sentences) reflecting on how you used the AI tool. Address the following:

- Which AI tool did you use and for which tasks?
- Did the AI make any mistakes? How did you catch and correct them?
- How did using AI affect your understanding of the material?
- What would you do differently next time?

General Instructions

Use the R Markdown report outline template you created in RStudio Cloud during lab to write a progress report addressing the tasks below. Knit your R Markdown code as a PDF report and submit it — including all code and answers — under "Data Analysis Project – Part I" in Blackboard by the due date.

💡 Tip: How AI Can Help with R Markdown

You can ask your AI tool: "What is R Markdown and how do I knit a PDF?" or "My R Markdown file won't knit — here is the error message. What should I check?"

Project Tasks

Task 1 — Explore the Dataset (3 points)

Use the glimpse() function from the tidyverse package to explore your assigned dataset. Answer the following:

a. How many cases (rows) are in your dataset?
b. How many variables (columns) are in your dataset?
c. Does your dataset contain missing values? Which variables contain them?

AI Prompt Idea for Task 1

"I used glimpse() on my dataset and got the following output: [paste output]. Can you help me understand what each part means?"

"How can I check for missing values in R? My dataset is called 'my_data'."

Task 2 — Research Questions

Identify at least two research questions you plan to answer using your dataset. It is strongly recommended to include at least one question answerable through visualization/correlation and at least one requiring predictive modeling.

2a. Research Questions & Hypotheses (6 points)

Write each research question and the corresponding hypothesis(es) to be tested.

AI Prompt Idea for 2a

"I am analyzing a dataset about [topic]. My variables include [list variables]. Can you help me brainstorm meaningful research questions that involve both visualization and prediction?"

"Here is my research question: [write it]. Help me formulate a testable hypothesis for it."

2b. Relevant Variables (2 points)

For each research question, list the relevant variables that will be used.

- Question 1: [list of relevant variables]
- Question 2: [list of relevant variables]
- Question 3 (if applicable): [list of relevant variables]

2c. Response Variable (2 points)

Identify the main response (dependent) variable in your data.

AI Prompt Idea for 2c

"In my dataset, I am trying to predict [outcome]. Which variable should be my response variable and why?"

2d. Summary Statistics for Response Variable (5 points)

Compute and report summary statistics (mean, standard deviation, and five-number summary) for the distribution of the response variable identified in 2c.

AI Prompt Idea for 2d

"Here is my R code for summary statistics: [paste code]. Can you explain what each number means in the context of my dataset?"

"What R functions can I use to compute mean, standard deviation, and the five-number summary?"

2e. Distribution Graphs (6 points)

Create at least three graphs displaying the distributions of three different variables, including the response variable. Use bar charts for categorical variables and histograms, dotplots, or boxplots for quantitative variables. Each graph must use a different variable.

AI Prompt Idea for 2e

"I want to create a histogram in R for my variable called 'income' using ggplot2. Can you show me the code structure?"
"My variable 'region' is categorical. What type of graph is best to show its distribution?"

✗ Do NOT do this

Do NOT ask: "Create three graphs for my MATH 224 project." and paste the resulting code without understanding it.
Do NOT use the same variable in more than one graph.

2f. Association Graphs (6 points)

Create at least two graphs showing the association between your response variable and two explanatory variables. Use scatter plots for quantitative explanatory variables and side-by-side boxplots for categorical explanatory variables.

AI Prompt Idea for 2f

"I want to show the relationship between house price (response) and lot size (quantitative explanatory variable) using ggplot2. What type of plot should I use and how do I make it?"
"How do I make a side-by-side boxplot in R to compare income across different education levels?"

Submission Checklist

- R Markdown knitted as a PDF with all code visible
- All six sub-tasks of Task 2 are addressed
- At least 3 distribution graphs (different variables each)
- At least 2 association graphs
- Copy and paste the entire AI conversation, in addition to the link, into a text doc (.txt)
- Reflection paragraph included at the end
